

PHYSICAL GEOGRAPHY



K. Allison Lenkeit-Meezan
Foothill College

Foothill College
Physical Geography

K. Allison Lenkeit-Meezan

This text is disseminated via the Open Education Resource (OER) LibreTexts Project (<https://LibreTexts.org>) and like the thousands of other texts available within this powerful platform, it is freely available for reading, printing, and "consuming."

The LibreTexts mission is to bring together students, faculty, and scholars in a collaborative effort to provide an accessible, and comprehensive platform that empowers our community to develop, curate, adapt, and adopt openly licensed resources and technologies; through these efforts we can reduce the financial burden born from traditional educational resource costs, ensuring education is more accessible for students and communities worldwide.

Most, but not all, pages in the library have licenses that may allow individuals to make changes, save, and print this book. Carefully consult the applicable license(s) before pursuing such effects. Instructors can adopt existing LibreTexts texts or Remix them to quickly build course-specific resources to meet the needs of their students. Unlike traditional textbooks, LibreTexts' web based origins allow powerful integration of advanced features and new technologies to support learning.



LibreTexts is the adaptable, user-friendly non-profit open education resource platform that educators trust for creating, customizing, and sharing accessible, interactive textbooks, adaptive homework, and ancillary materials. We collaborate with individuals and organizations to champion open education initiatives, support institutional publishing programs, drive curriculum development projects, and more.

The LibreTexts libraries are Powered by [NICE CXone Expert](#) and was supported by the Department of Education Open Textbook Pilot Project, the California Education Learning Lab, the UC Davis Office of the Provost, the UC Davis Library, the California State University Affordable Learning Solutions Program, and Merlot. This material is based upon work supported by the National Science Foundation under Grant No. 1246120, 1525057, and 1413739.

Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the National Science Foundation nor the US Department of Education.

Have questions or comments? For information about adoptions or adaptations contact info@LibreTexts.org or visit our main website at <https://LibreTexts.org>.

The LibreTexts name, logo, and book covers are trademarked by LibreTexts, Inc. (a 501(c)(3) not-for-profit organization) and protected against unauthorized use.

This text was compiled on 07/19/2026

- [InfoPage](#) by [Delmar Larsen](#) is licensed [Public Domain](#).

TABLE OF CONTENTS

Licensing

1: Measuring, Monitoring and Describing the Earth

- 1.1: Cartography
- 1.2: Location and Time on Earth
- 1.3: Map Scale and Projection
- 1.4: Remote Sensing
- 1.5: Geographic Information Systems (GIS)

2: Weather and Climate

- 2.1: Air Masses and Weather
- 2.2: Violent Weather
- 2.3: Global Climate Patterns

3: Earth Materials - The Lithosphere

- 3.1: The Earth's Structure
- 3.2: Geologic Time
- 3.3: Plate Tectonics
- 3.4: The Rock Cycle

4: Shaping the Lithosphere

- 4.1: Volcanoes
- 4.2: Volcanic Hazards
- 4.3: Orogenesis, Faults and Earthquakes

5: The River and the Sea

- 5.1: Fluvial Processes
- 5.2: Stream Transport
- 5.3: Flood Management
- 5.4: Where River Meets Sea - The Coastal Environment
- 5.5: The Oceans

6: Quizzes and Exams

- 6.1: Quizzes
- 6.2: Exams

Index

Glossary

Detailed Licensing

Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Measuring, Monitoring and Describing the Earth

Learning Objectives

By the end of this chapter you should be able to

- List the ways in which Geographers use maps
- Demonstrate how longitude relates to time on earth
- Describe how scale and map projection can be used to distort ideas
- Relate electromagnetic radiation to remote sensing
- Define a Geographic Information System

[1.1: Cartography](#)

[1.2: Location and Time on Earth](#)

[1.3: Map Scale and Projection](#)

[1.4: Remote Sensing](#)

[1.5: Geographic Information Systems \(GIS\)](#)

This page titled [1: Measuring, Monitoring and Describing the Earth](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

1.1: Cartography

How do we know exactly where we are? How do we represent our world? Cartography is the science of map making. Maps are representations of the world, usually two-dimensional, graphic relationships that use lines, symbols and colors to convey information or ideas about spatial relationships. Geographers use maps for four reasons:

1. *To locate phenomena*- such as places, disease, peoples, etc.
2. *To show relationships* - using maps to prove that some traits are correlated, such as population growth and infant mortality rates, or air pollution and auto usage.
3. *To prove ideas* - maps illustrate relationships (as mentioned above) and statistics which allow us to support our ideas with location examples
4. *To ask questions* - at times maps may reflect areas of contradiction from basic relationships. Such anomalies enable us to question the causal factors we are addressing.

These are four thematic maps of the United States. Each communicates different information, each could be used for different purposes. The most familiar maps are a combination choropleth (where colors have values) and vector (line) maps. Your standard road atlas is also usually a combination choropleth and vector map, where colors represent land cover (green for forested areas, blue for water, and sometimes yellow for urban areas), and roads are represented by lines. A third type of map is an isopleth map, where lines represent changes of a continuous value. A USGS topographic map is an example of an isopleth map.

In a standard choropleth map, the colors represent value added by manufacture. This is a graduated color scheme, commonly used to convey a range of values. Without even looking at the legend in this graduated color scheme choropleth map, you can probably figure out that the darker colors (reds and dark oranges) represent high values, and the lighter colors represented lower values.

Another type of map uses pie charts to convey labor structures. While the divisions of the pie chart represent the types of labor in each city, the *size* of the pie chart represents the total size of the labor force. In this way, the map is conveying both the distribution of the labor force by *type* and also by *size*. Such a map would answer questions like: Where is the majority of the labor force in the US located? Where is the labor force not located? What reasons can you think of for this distribution? These are the types of questions that a good map should prompt you to contemplate.

Browse through more examples of maps.

This page titled [1.1: Cartography](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

1.2: Location and Time on Earth

We define where we are on earth by several grid systems. The most common of these is **latitude and longitude**. Latitude and longitude divide the earth up into a series of parallel lines (latitude) and vertical lines running from the north pole to the south pole (longitude). Your book has a very good description of latitude and longitude, so read it carefully.

The base lines for measuring latitude and longitude are the **equator** and the **prime meridian** (see the graphic below). The equator goes around the middle of the earth, or it is the longest circumference. The prime meridian is an arbitrary line of longitude, but it was picked to go through Greenwich, England, the location of the Royal Observatory. Latitude is measured from 0 to 90 degrees north of the equator (0 to 90N), and 0 to 90 degrees south of the equator (0 to 90S), with 0 being the equator. It is easy to remember this by thinking about 180 degrees in a hemisphere, or half circle. Longitude is measured from 0 to 180 degrees east of the Prime Meridian (0 to 180E) and 0 to 180 west of the Prime Meridian (0 to 180W). Therefore, everything in the United States has a northerly latitude (*it is in the northern hemisphere*), and a westerly longitude (*it is in the western hemisphere*).

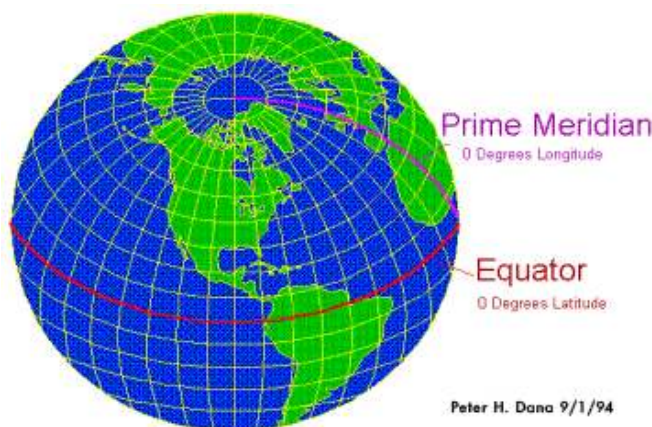


Figure 2.3.1 Latitudes and Longitudes

Increments of latitude and longitude are called 'degrees'. One degree is divided into 60 minutes ('). One minute is divided into 60 seconds ("). So a location might look something like this: 37°15' 23"N, 122°14'8"W (where is this?).

Conduct some research about the relationship between time and longitude. Each 15° increment in longitude equals one hour of time on earth. We have roughly divide the earth up in this manner by time zones. You can therefore roughly estimate (ignoring daylight savings time and the peculiarities of political gerrymandering of time zones) how many hours earlier or later one location is from another. For example, if you were in a ship at 20°W and you sailed to 80° W, what time would it be in your new location if it was 6pm in your old location? You would have traveled 60° to the west, so $60/15 = 4$, and west equals earlier, so it would be 2pm in your new location.

Check out the web site World Clock for time by location on earth.

This page titled [1.2: Location and Time on Earth](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

1.3: Map Scale and Projection

Map scale is defined as: *The ratio of distance on a map to distance on the ground* Map scale is generally expressed as a ratio, such as 1:100,000. This means that one unit on the map is equal to 100,000 units on the ground, or the map representation of a widget is 1/100,000th of the actual size of the widget.

One way that I find useful to visualize map scale is to think about a map of the world. The length of the equator at different scales is a good way to think about the actual size of a map at that scale. The table below lists the distance on the map (as if you were to lay a ruler along the map and measure the equator) for each map scale. As you can see, a 1:400,000,000 scale map would probably fit across two pages of your textbook, while a 1:10,000,000 scale map would take up a wall of your classroom. At 1:1,000 scale, a map of the world would stretch across the county!

The Length of the Earth's Equator at Different Map Scales

Map Scale	Distance of the Equator on the Map (m)
1:400,000,000	0.10002
1:40,000,000	1.0002
1:10,000,000	4.0008
1:1,000,000	40.008
1:100,000	400.078
1:10,000	4000.78
1:1,000	40,007.8

Large Scale Maps vs. Small Scale Maps

“Large Scale” <i>Small features are large</i> • A map of this room • A map of this campus • A map of this city	“Small Scale” <i>Large features are small</i> • A map of this country • A map of this world
---	---

Resolution is defined as: *The smallest feature that is represented on the map*

- A city such as Los Altos Hills probably **would not** be represented on a map of the USA (at a scale 1:10,000,000). The **resolution** of this map is therefore too coarse to represent our city.
- A city such as Los Altos Hills probably **would** be represented on a map of the Bay Area (scale 1:500,000). The **resolution** of this map is therefore fine enough to represent our city.

Map projection is the way in which we represent the spherical earth on a flat map (see below). You may want to do some searches to find different types of map projections. The most important thing to remember about map projections, however, is that *there will always be some distortion*. Some map projections preserve relative areas, some projections preserve shape (such as the shape of coast lines), some try to do both and end up doing neither. But the result is that there will *always* be some error in a projected map.

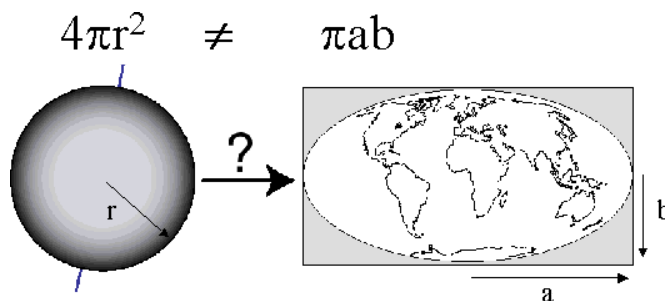


Figure 2.4.1 Map projection

One example of error in map projection can be seen in maps of the United States. The graphic below shows three different projections of the United States overlain. Note that the further you get from the center of the projection (the hot pink dot in the middle of the figure below) the bigger the distortion gets. Look in particular at Florida, Washington and Maine.

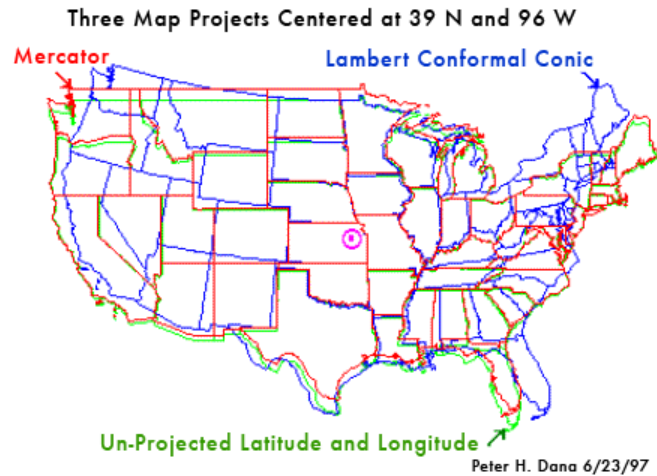


Figure 2.4.2 Map projection of the United States

Another example of distortion caused by map projection is the represented area of Australia and Greenland. The landmass of Greenland is 1/3 that of Australia. That is to say, Australia is three times as big as Greenland. The map on the left is a Mollweide Equal Area projection which preserves the relative area of landforms. The projection on the right is a Gall projection (very similar to a Mercator projection), and it preserves the shape of landforms but not the area.

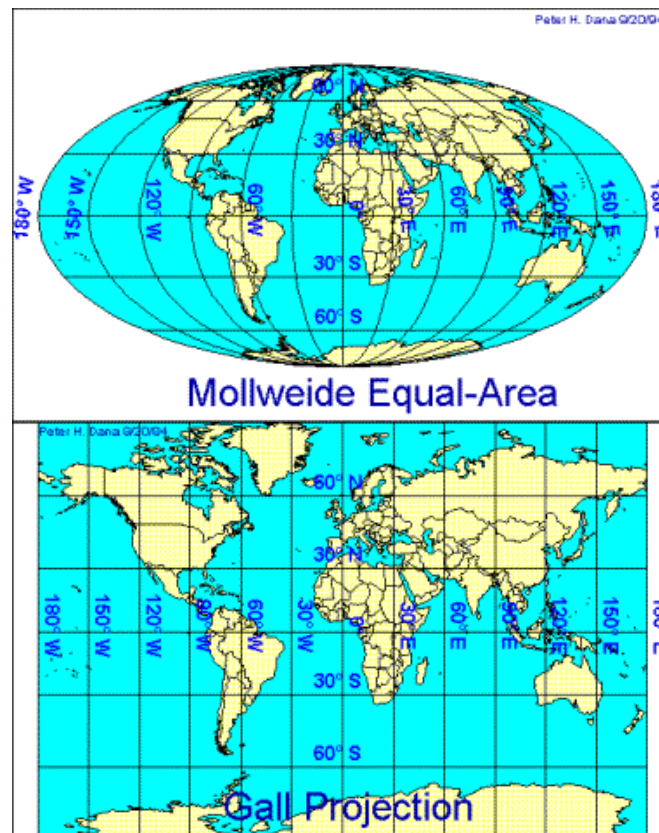


Figure 2.4.3 More map projections

This page titled [1.3: Map Scale and Projection](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

1.4: Remote Sensing

Remote sensing is very important to physical geography. It allows geographers to look at change over a large area over time. In addition, remote sensing takes advantage of the interaction of earth materials with various wavelengths of the electromagnetic spectrum (recall last week's lecture). The chemical properties of some materials preferentially absorb or reflect certain wavelengths. For example, the iron in chlorophyll (the 'energy' producing element of all green plants) reflects very strongly in the near infrared (the wavelength slightly longer than red, and just outside the range of human vision). Chlorophyll also absorbs very strongly in the blue wavelengths. Therefore by looking at the near infrared wavelength and comparing it with the blue, you can pick out the vegetated areas by looking for areas that reflect strongly in red, and not in blue.

The following web pages are excerpts from a NASA tutorial on remote sensing. Some of the material is pretty technical, but read it carefully, and pay particular attention to the graphs:

- Electromagnetic Spectrum: Spectral Signatures
- Processing and Classification of Remotely Sensed Data

In the first graph, which compares Pinewood, Grassland, Red Sand Pit and Silty water, if you were looking at a remotely sensed image at 0.8 micrometers wavelength, which would you expect to appear the brightest? Which would appear the least bright?

The Grasslands would appear the brightest, as it reflects the most electromagnetic energy at that wavelength. The silty water would appear the darkest, as it would reflect the least at that wavelength. An example of this type of remote sensing is the Normalized Difference Vegetation Index or NDVI. NDVI is a ratio of near infrared wavelengths to visible wavelengths, and can be thought of as 'vegetation vigor'. The figure below is an example of global NDVI for January and July. Recall that December and June are the solstices or the height of summer or winter. The figures below are selected as January and July because vegetation generally has a lag-time between when the conditions are at a maximum (height of summer or winter) and when they are at their greenest. In the figure below, light colors indicate low vegetation vigor, while dark colors indicate high vegetation vigor.

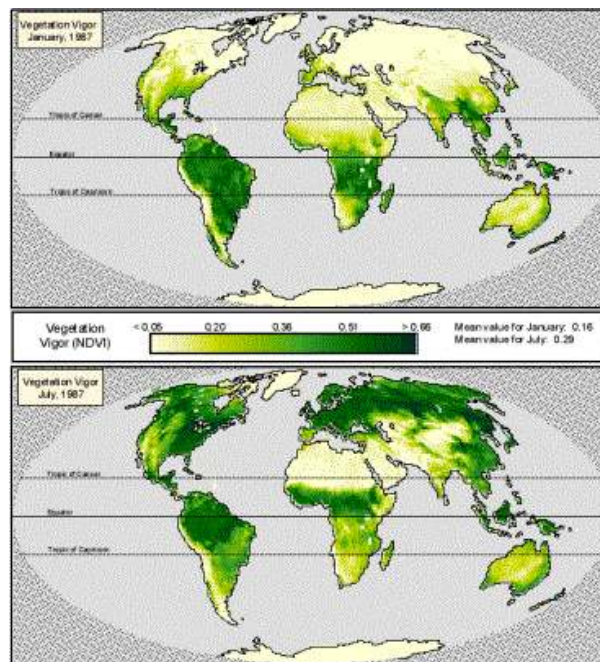


Figure 2.5.1 Global NDVI for January and July 1987

In addition to optical remote sensing (remote sensing that uses the visible and near infrared portion of the electromagnetic spectrum), another common type of remote sensing is radar. Radar remote sensing measures the 'roughness' of a surface. It is very useful for studies which look at land-use change because man-made features tend to be very smooth compared to vegetation which is rough. Think about if you were in an airplane looking down at a city and a forest. The forest looks 'fuzzy' and 'soft' while the buildings and roads of the city have large flat, 'smooth' surfaces. See a radar image of the San Francisco Bay Area. Note that the urban areas (smooth) reflect very strongly (have a bright color) and the vegetated hills surrounding the Bay Area (rough) do not reflect (appear dark).

This page titled [1.4: Remote Sensing](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

1.5: Geographic Information Systems (GIS)

A geographic information system (GIS) is a suite of map layers or ‘themes’ that are all georeferenced to the same location on earth, and their associated tabular spreadsheet data. A GIS is a map, a spreadsheet and an information management system all rolled into one. Conduct an internet search for a common rendition of a GIS. The map layers of a GIS are all stacked on top of one another in the computer, allowing the user to query all the layers at once. GIS are used in numerous professional fields, including Geography, Archaeology, Biology, Business, Forestry, Geology, Emergency Management and Law Enforcement.

The best way to experience GIS is to experiment with a GIS. Go to Ice Maps , at the UC Davis Information Center for the environment. Start out by selecting several layers to view. Some suggested combinations are listed below. Look at the legend, zoom in and out and pan around the state then discuss the following prompt in the DC.

Suggested theme combinations:

- Shaded Relief, Major California Rivers, Lakes, Jurisdictional Dams, Non Jurisdictional Dams
- Shaded Relief, Major Cities, State Highways, Federal Highways, Proposed Significant Natural Areas

This page titled [1.5: Geographic Information Systems \(GIS\)](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

CHAPTER OVERVIEW

2: Weather and Climate

Learning Objectives

By the end of this chapter you should be able to

- Describe at least three mechanisms that will cause clouds to form
- Discuss the effects of violent weather such as hurricanes and tornadoes on people
- Apply geographic principles of climate introduced in class to predict climate in Modelland

[2.1: Air Masses and Weather](#)

[2.2: Violent Weather](#)

[2.3: Global Climate Patterns](#)

This page titled [2: Weather and Climate](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

2.1: Air Masses and Weather

In chapter 3, we discussed how warm air had more energy, and therefore formed areas of low pressure in the atmosphere, while cold air was more dense and formed areas of high pressure in the atmosphere. In chapter 4, we discussed atmospheric and oceanic circulation and global temperatures. In the previous lesson, we talked about the water cycle and water resources on earth. In this unit, we are going to bring together these discussions about the atmosphere/hydrosphere interactions in our exploration of weather and global climate.

First, let's get some very important definitions out of the way. Weather and climate are not the same thing. *Weather* is the short-term, day to day condition of the atmosphere, while *climate* is the long term (over decades) average of weather conditions in a region. Climate change can only be documented by noting a rise the average annual global temperature over many years. Another way to think about is that weather is a series of short-term events, while climate is a long-term trend. A person who studies and predicts the weather is called a *meteorologist*. A person who studies long-term trends in weather and how climate changes over many years is called a *climatologist*.

Air masses are classified based on their moisture and temperature. Air masses pick up moisture when they are over bodies of water, and they lose that moisture when they are over land. Therefore, wet air masses are designated **m** for maritime, and dry air masses are designated **c** for continental. The temperature of an air mass is designated as follows: **A** for arctic, **P** for polar (high latitude, but not as high as arctic), **T** for tropical, **E** for equatorial and **AA** for Antarctic.

The following air masses affect the United States. Write down their abbreviations in the parenthesis below:

- Maritime Polar (___) affects the north west and north east coasts all year.
- Continental Polar (___) affects the north central USA in the summer and more severely in the winter.
- Maritime Tropical (___) affects the south west and south east coasts all year.
- Continental Tropical (___) affects the south central US during the summer.

Clouds and violent weather occur when air masses collide. When a cold mass pushes under a warm air mass, the warm air is cooled adiabatically (recall the average lapse rate of 6.4 degrees C per kilometer from lecture 3). The moist warm air quickly reaches its dew point and clouds form. If you watch the weather report on the evening news, you will note that the cold fronts are denoted by lines with triangles pointing in the direction that the front is advancing. Warm fronts are denoted by lines with half-circles pointing in the direction that the front is advancing.

Another important atmospheric lifting mechanism is *orographic* lifting. Orographic lifting occurs when air is forced up to high altitudes as it is pushed up a mountain front. The air cools adiabatically, and clouds form, and precipitation often results. The windward side of the mountain, therefore, receives much more precipitation than the leeward side. This is known as the '*rainshadow effect*'. One example of orographic lifting is the Sierra Nevada mountains. Moist air blows off of the Pacific ocean and is forcibly lifted upslope. This results in significant rain and snow on the western side of the Sierras (Tahoe, for example), and very little of the precipitation makes it over to the east side of the Sierras. This results in a dry desert on the eastern slope. If you have ever driven over highway 80 to Reno, or highway 120 past Yosemite, you will have seen this transformation first hand. The western side of the Sierras boast lush forests, while the eastern slope is a dry, sage brush desert.

This page titled [2.1: Air Masses and Weather](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).

2.2: Violent Weather

Two types of violent weather affect the United States each year: **tornados and hurricanes**. The U.S. has more tornadoes each year than any other country in the world. There are two conditions necessary for tornadoes to form. First, fast winds in the upper troposphere are slowed at the ground because of friction. This condition, of fast winds in the upper troposphere, and slower winds in the lower troposphere, causes a tube of horizontally rotating air to form. For a tornado to form, something must make that column of rotating air turn vertical. A strong updraft of rising hot air can be powerful enough to twist the horizontally rotating column of air into a vertical column of low pressure air.

Thunderstorms that precede tornadoes usually form in the late afternoon after the sun has warmed the ground. The air heated over the ground is buoyant and it rises and rapidly reaches the dew point causing the formation of massive cumulonimbus clouds. The updraft of the rising air is often enough to turn the horizontally rotating air in the troposphere into a vertical funnel cloud. In the late 1960's, Dr. T. Theodore Fujita, an authority on tornadoes at the University of Chicago, proposed a scale (called the Fujita scale) for classifying tornadoes according to their rotational wind speed and the damage done by the storm.

Browse this web page for more information:[NOAA FAQ about Tornadoes](#)

Tropical cyclones are another awesome expression of nature's power. Tropical cyclones go by several different names, depending on their location. *Hurricane* is the term used in the Atlantic and East Pacific, *Typhoon* is the term used in the Western Pacific, and *Cyclone* is the term used in the Indian Ocean or near Australia. Cyclones are zones of intense low pressure. The Atlantic basin cyclones that you investigated in Lab 3 generally started on the western coast of Africa as zones of low pressure resulting from superheated land. These low pressure zones then moved out over the central Atlantic and gathered the abundant energy stored as latent heat in the warm water. By the time that they reached the eastern coast of the Americas they were fully developed, powerful cyclones.

Browse this NOAA page on Hurricanes for more information.

Note that tornadoes can be a bi-product of hurricanes!

This page titled [2.2: Violent Weather](#) is shared under a [CC BY-NC 4.0](#) license and was authored, remixed, and/or curated by [K. Allison Lenkeit-Meezan](#).